

Hala Alzureiqi

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Education

University of Waterloo

Expected May 2030

Bachelor of Applied Science – Biomedical Engineering, Honours, Co-operative Program

84% Cumulative Average

- **Relevant Coursework:** Data Structures & Algorithms (C++), Introduction to Chemistry, Engineering Biology, Human Factors in Biomedical & Health Systems

Experience

Research Intern – Trust Management in Human-Autonomy Teaming

May 2026 – Aug. 2026

National Research Council Canada, Digital Technologies Research Centre

Ottawa, ON

- Conducted literature reviews on trust measurement in human-autonomy teaming to design 2 trust-violation scenarios, author their experimental protocol, and conduct pilot studies to detect indicators of trust loss
- Integrated and calibrated non-invasive sensors (GSR, eye tracking, facial expressions, voice) using iMotions to capture physiological and behavioural responses across 75 participants following REB approval
- Developed Python tools for multimodal sensor data preprocessing and analysis and evaluated multiple machine learning approaches for classifying changes in operator trust

Electrical Sub-team Member

Oct. 2025 – Present

UW Biomedical Engineering Society

Waterloo, ON

- Researched analog filtering techniques for EMG signal acquisition and supported circuit testing
- Programmed ESP32s to stream multi-channel EMG over BLE via custom GATT services for an assistive knee brace
- Built a Python BLE client with a GUI to stream, filter, and visualize live EMG signals; validated against expected muscle activation patterns and supported device testing

Tech Lead

Jan. 2026 – Present

UW Muslim Student Association

Waterloo, ON

- Evaluated approaches for a private internal AI search tool designed to preserve MSA institutional knowledge and improve access to information for current and future executives
- Developed the vector database component of the RAG pipeline using PostgreSQL and pgvector, supporting ingestion of PDF, DOCX, and XLSX documents for citation-backed question answering
- Worked with executive team members to identify information-access needs and translate them into technical requirements for the system

Skills & Interests

- **Laboratory Techniques:** WHMIS, pipetting, solution preparation and dilution, titration, microscopy, contact-angle measurement, alginate hydrogel preparation, spectrophotometry
- **Programming & Data Analysis:** Python (NumPy, SciPy, PyTorch, OpenCV), MATLAB, C++, Git
- **Instrumentation & Testing:** iMotions, GSR, eye tracking, EMG, ESP32, BLE, sensor integration, force testing
- **CAD & Prototyping:** SolidWorks, 3D printing, tolerance analysis, CMM 3D scanning

Projects

Inclusive Catan Redesign | *SolidWorks, 3D Printing*

Sep. – Dec. 2025

- Calculated magnetic and friction forces analytically to determine a 24.9 N tile-separation force and 17.8 N grip force, validating the design against QFD thresholds
- Designed and prototyped a magnetically connected, modular game board and stabilized playing pieces using SolidWorks and 3D-printed PLA, improving stability while preserving original gameplay mechanics
- Conducted quantitative force testing with a calibrated force gauge, measuring up to 58 N to displace settlements, exceeding the 30 N QFD stability target